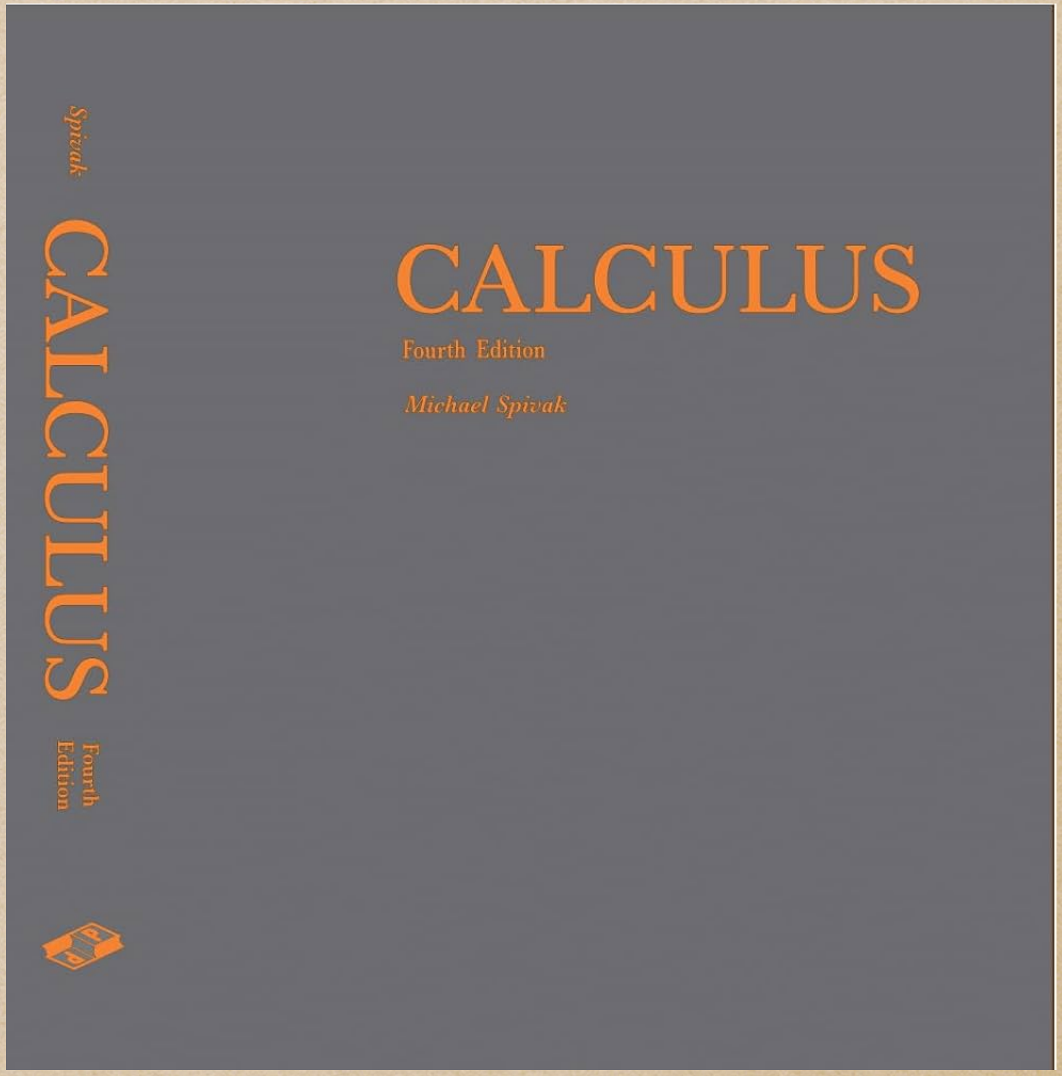
Week 6.1

Definite Calculus

Definite Calculus

- Definite Integrals
- Definite Derivatives
- MATLAB sorcery



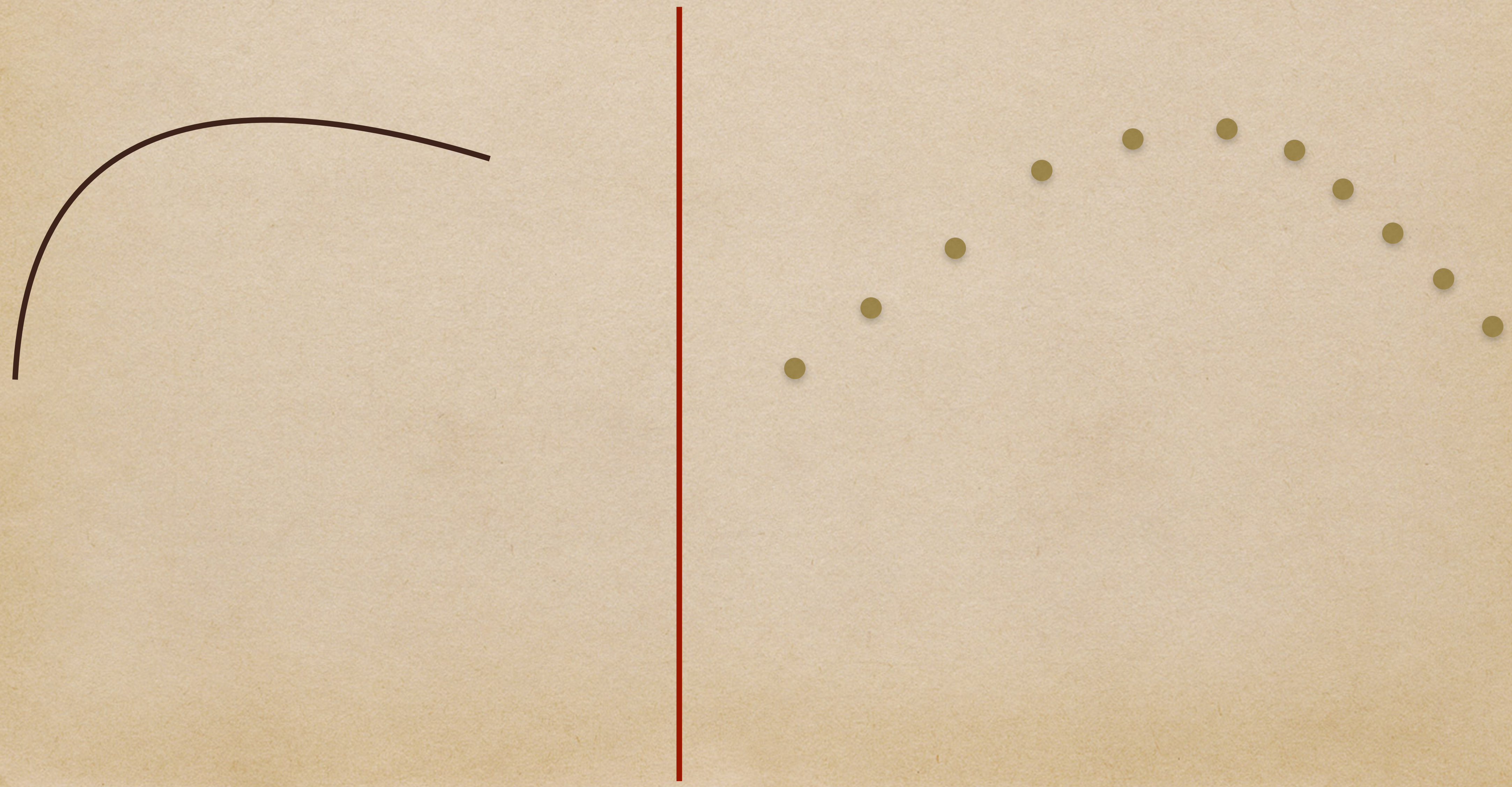
Dr. Michael Spivak

Definite Integrals

- What is an integral?
- Why is an integral useful?
- How does one integrate?

Definite Integrals

- Continuous space vs Discrete space



Area under curve

- Riemann method(s)
- Trapezoid Rule
- Simpson method (out of scope 😭)

Riemann methods

- Squares
- Arbitrary interval

Trapezoid Rule

- Trapezoids
- Arbitrary intervals
- Avg of Riemann right and Riemann left

“ Most of us know the law of the lever, but this law is simply a quantitative statement of exactly how amazing the lever is, and doesn't give us a clue as to why it is true, how such a small force at one end can exert such a great force at the other.”

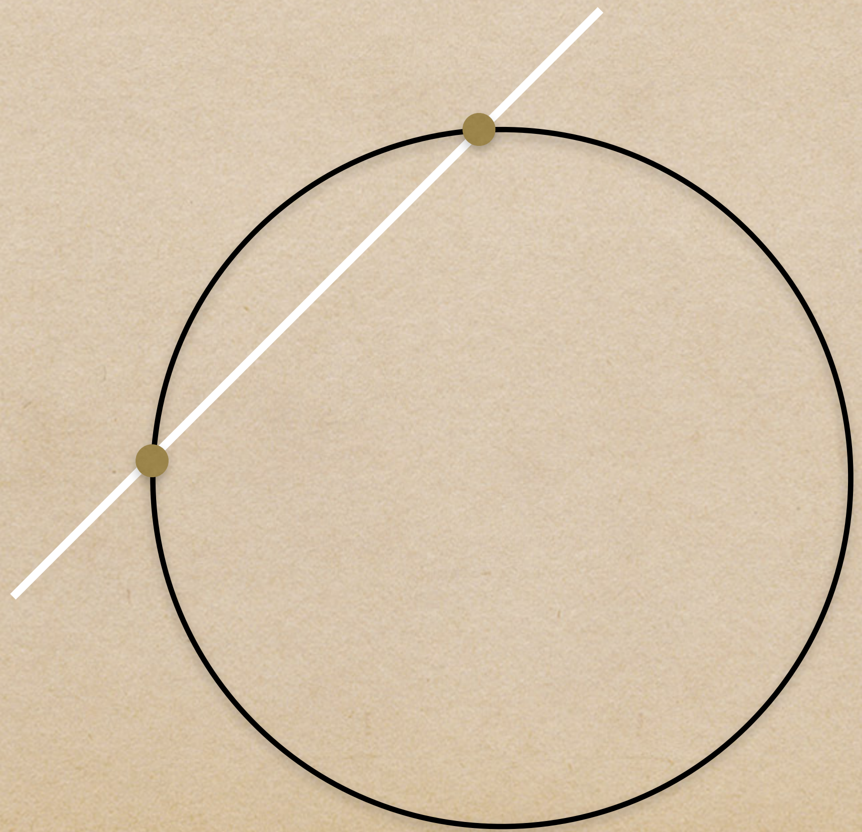
- Michael Spivak PhD.

Definite derivative

- What is a derivative?
- Why is a derivative useful?
- How does one get a derivative (by hand)?

Difference quotient

- secant line $\dfrac{f(x_a) - f(x_b)}{\Delta x}$ s.t $\Delta x = x_b - x_a$



MATLAB sorcery

- *int*()
- *diff*()

Questions?